

Mini Project 01 – JpetStore

Introduction

In this JpetStore built by OctoPerf, which is used to purchase pets like fish, dogs, cats, reptiles, and birds. We are going to perform the load test, stress test, spike test, endurance test, and volume test to be here.

- JPetStore web application at <https://www.blazedemo.com/>
- **Performance goals** response time will be 3 secs.
- Error rate should remain **< 2% during normal load and < 5% under stress** conditions.

Test Setup

OctoPerf

1. [JPetStore Demo \(octoperf.com\)](https://petstore.octoperf.com/) – Home Page
2. <https://petstore.octoperf.com/actions/> - Choose Cart
3. [JPetStore Demo \(octoperf.com\)](https://petstore.octoperf.com/) - Add cart
4. [JPetStore Demo \(octoperf.com\)](https://petstore.octoperf.com/) – Select Quantity
5. [JPetStore Demo \(octoperf.com\)](https://petstore.octoperf.com/) - Check Out

Test-Setup

1. From Recording controller to get the data from website.
2. Using Synchronizing Timer to integrate n no of user requests.
3. Using Random Controller to process the requests on randomly.
4. Using Loop Controller in infinite mode to repeat the activity.
5. The above features is integrate with normal thread group by duration to stop specify thread lifetime according to the test case.

Environment Setup

1. **Processor** - 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz 2.42 GHz
2. **RAM** – 8GB

Jenkins

It integrated with Jenkin's by running with the local environment.

GitHub

1. [Performance_Testing/Pet_Shop_\(Load_Test\).jmx at master · senthilraja29/Performance_Testing \(github.com\)](https://github.com/senthilraja29/Performance_Testing/blob/master/Performance_Testing/Pet_Shop_(Load_Test).jmx)
2. [Performance_Testing/Pet_Shop_\(Stress_Test\).jmx at master · senthilraja29/Performance_Testing \(github.com\)](https://github.com/senthilraja29/Performance_Testing/blob/master/Performance_Testing/Pet_Shop_(Stress_Test).jmx)
3. [Performance_Testing/Pet_Shop_\(Spike_Test\).jmx at master · senthilraja29/Performance_Testing \(github.com\)](https://github.com/senthilraja29/Performance_Testing/blob/master/Performance_Testing/Pet_Shop_(Spike_Test).jmx)
4. [Performance_Testing/Pet_Shop_\(Scalability_Test\).jmx at master · senthilraja29/Performance_Testing \(github.com\)](https://github.com/senthilraja29/Performance_Testing/blob/master/Performance_Testing/Pet_Shop_(Scalability_Test).jmx)
5. [Performance_Testing/Pet_Shop_Endurance_Test_recent.jmx at master · senthilraja29/Performance_Testing \(github.com\)](https://github.com/senthilraja29/Performance_Testing/blob/master/Performance_Testing/Pet_Shop_Endurance_Test_recent.jmx)

Test Results

Load Test

Latency Report

Latency				
Row Labels	StdDev of Latency	Average of Latency	Max of Latency	Min of Latency
Billing_Address	672.8810543	719.4422982	35852	150
Billing_Form	812.1504933	732.3654105	35877	147
Cart	630.5111981	715.5459587	35864	148
Choose_Pet	722.4803047	750.0991597	35863	148
Delivery_Address	713.8977491	719.5807325	35868	149
Home Page	973.0502511	657.6331948	35873	148
Pet_Speacification	852.3118581	751.3677232	35877	149
Product Page	806.3320059	731.7489299	35864	147
Select_Product	788.698715	744.2373374	35869	149
Signin_Page	825.6668631	751.0474463	35877	149
Signin_Progress	759.4159945	737.9700874	35863	150
SignOut	360.5138481	492.1721301	35860	148
SignOut-0	360.5138481	492.1721301	35860	148
SignOut-1	995.5955916	324.5630606	35288	147
Grand Total	766.2454012	665.3306035	35877	147

- 1. Average Latency:** The overall average latency is approximately 665 ms. The "Choose_Pet" step has the highest average latency (750 ms), while "SignOut-1" shows the lowest average latency (324 ms).
- 2. Standard Deviation (StdDev):** The average StdDev is 766 ms, with "SignOut-1" exhibiting the highest variability (995 ms), indicating inconsistent performance.
- 3. Maximum Latency:** All steps have maximum latency peaks around 35,800 ms, showing occasional long delays across processes.
- 4. Minimum Latency:** Most steps have minimum latencies between 147 ms and 150 ms, except for "SignOut-1" (147 ms).

Aggregate Report

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughp...	Received ...	Sent KB/sec
Home Page	24070	657	702	989	1297	2104	148	35873	0.00%	13.4/sec	18.96	7.23
Choose_Pet	23921	750	855	1107	1315	1957	148	35863	0.00%	13.3/sec	47.59	8.79
Delivery_...	23646	785	858	1138	1409	1947	149	35868	0.00%	13.1/sec	57.96	15.06
Signin_Pa...	23711	753	854	1109	1318	1973	149	35877	0.00%	13.2/sec	52.69	9.03
Select_Pro...	24067	744	853	1112	1329	1935	149	35869	0.00%	13.4/sec	49.90	9.26
Billing_Fo...	23776	801	862	1153	1406	2017	147	35877	0.00%	13.2/sec	54.46	8.91
Cart	24163	784	859	1143	1409	1962	148	35864	0.00%	13.4/sec	61.23	9.22
Pet_Speac...	24135	751	853	1125	1352	2042	149	35877	0.00%	13.4/sec	49.73	9.19
Product P...	24296	800	860	1136	1407	2067	147	35868	0.00%	13.5/sec	65.56	8.23
Signin_Pr...	24137	761	855	1131	1401	2114	150	35863	0.00%	13.4/sec	54.68	13.83
Billing_Ad...	24280	787	858	1139	1403	1943	150	35852	0.00%	13.5/sec	60.23	17.89
SignOut	24191	876	884	1133	1358	1929	297	36252	0.00%	13.4/sec	69.26	17.07
TOTAL	288393	771	851	1123	1375	1999	147	36252	0.00%	160.2/sec	642.06	133.66

1. **Average Response Time:** The total average response time is **771 ms**. "SignOut" has the highest average response time of **876 ms**, while "Home Page" has the lowest at **657 ms**.
2. **Throughput:** All processes have a steady throughput of **13.4 requests/sec**, except the total throughput, which is significantly higher at **160.2 requests/sec**.
3. **Error Rate:** All steps show a **0.00% error rate**, indicating no failures during the tests.
4. **Response Percentiles:** The **95% line** indicates most requests complete in under **1392 ms**, with a few outliers in the **99% line** exceeding **1999 ms**.
5. **Data Transfer:** "SignOut" transfers the most data with **69.26 KB/sec received** and **21.58 KB/sec sent**.

Summary Report

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB...	Sent KB/sec	Avg. Bytes
Home Page	24070	657	148	35873	973.03	0.00%	13.4/sec	18.96	7.23	1452.0
Choose_Pet	23921	750	148	35863	722.71	0.00%	13.3/sec	47.59	8.79	3666.6
Delivery_Ad...	23646	785	149	35868	724.28	0.00%	13.1/sec	57.96	15.06	4518.2
Signin_Page	23711	753	149	35877	825.10	0.00%	13.2/sec	52.69	9.03	4095.7
Select_Prod...	24067	744	149	35869	788.89	0.00%	13.4/sec	49.90	9.26	3820.8
Billing_Form	23776	801	147	35877	852.97	0.00%	13.2/sec	54.46	8.91	4222.5
Cart	24163	784	148	35864	681.83	0.00%	13.4/sec	61.23	9.22	4669.9
Pet_Speacifi...	24135	751	149	35877	852.46	0.00%	13.4/sec	49.73	9.19	3797.6
Product Page	24296	800	147	35868	875.00	0.00%	13.5/sec	65.56	8.23	4973.5
Signin_Prog...	24137	761	150	35863	762.52	0.00%	13.4/sec	54.68	13.83	4175.6
Billing_Addr...	24280	787	150	35852	755.44	0.00%	13.5/sec	60.23	17.89	4572.1
SignOut	24191	876	297	36252	1133.67	0.00%	13.4/sec	69.26	17.07	5278.1
TOTAL	288393	771	147	36252	839.29	0.00%	160.2/sec	642.06	133.66	4104.7

1. **Overall Performance:** The system demonstrates consistent performance across all pages, with average response times ranging from 657 to 876 milliseconds.
2. **Throughput:** The system is capable of handling an average of 13.1 to 13.5 requests per second

Stress Test

Latency Report

Latency				
Row Labels	StdDev of Latency	Average of Latency	Max of Latency	Min of Latency
Billing_Address	151.1652957	222.8884436	1727	0
Billing_Form	390.7655603	552.8500665	3509	150
Cart	381.3621133	542.0262677	2381	148
Choose_Pet	125.892157	196.9591005	3656	146
Confirmation	151.4224104	222.1432144	1725	148
Home Page	100.1761209	186.9783308	1306	146
Pet_Speacification	147.7359118	210.7243579	2047	147
Product Page	102.2553028	191.8256881	1236	147
Select_Product	111.9314346	194.8806094	1157	147
Signin_Page	111.9164483	194.2161681	1519	146
Signin_Progress	163.6518443	226.040382	1860	149
SignOut	128.1525274	184.383738	7680	145
SignOut-0	128.1525274	184.383738	7680	145
SignOut-1	74.62001923	170.3912335	3311	147
Grand Total	223.7868388	248.2024017	7680	0

- Overall Latency:** The system exhibits an overall average latency of 248 milliseconds.
- Variability:** "Billing_Address" and "Cart" show the highest variability in latency, as indicated by their standard deviations. This suggests that these pages may experience significant fluctuations in response times.
- Extremes:** "Sign Out" has the highest maximum latency (7680 ms), indicating a potential outlier or bottleneck. In contrast, "Billing_Address" has the lowest minimum latency (0 ms), suggesting a consistently fast response time

Aggregate Report

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Through...	Received ...	Sent KB/s...
Billing_Form	6016	634	531	1214	1274	1643	150	3614	0.00%	10.1/sec	41.61	6.80
Cart	6015	621	516	1203	1261	1514	148	3883	0.00%	10.1/sec	46.00	6.91
SignOut	6137	370	325	495	640	851	293	7885	0.00%	10.3/sec	53.07	13.08
Select_Product	6039	194	163	190	483	697	147	1157	0.00%	10.1/sec	37.82	7.01
Home Page	5907	186	160	175	475	652	146	1306	0.00%	9.9/sec	14.05	5.36
Signin_Page	6185	196	162	304	483	700	146	1519	0.00%	10.4/sec	41.52	7.11
Choose_Pet	6137	196	162	208	485	701	146	3656	0.00%	10.3/sec	36.90	6.80
Product Page	5995	209	163	332	485	674	147	1348	0.00%	10.1/sec	48.87	6.12
Pet_Speacification	6113	210	165	382	498	883	147	2047	0.00%	10.2/sec	38.04	7.02
Signin_Progress	5968	238	174	487	550	1045	149	2191	0.00%	10.0/sec	40.77	10.30
Confirmation	5998	247	175	483	526	1053	148	1918	0.00%	10.0/sec	44.32	11.50
Billing_Address	5988	250	175	486	531	1029	148	21037	0.02%	10.0/sec	44.79	13.29
TOTAL	72498	296	168	657	988	1264	146	21037	0.00%	121.0/sec	485.81	100.90

Response Time Analysis Summary

1. Average Response Times:

Highest: Billing Form (634 ms)

Lowest: Select Product (194 ms)

2. 95th Percentile Response Times:

Highest: Billing Form (1214 ms)

Lowest: Home Page (345 ms)

3. 99th Percentile Response Times:

Highest: Billing Form (1643 ms)

Lowest: Home Page (652 ms)

Summary report

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB...	Sent KB/sec	Avg. Bytes
Billing_Form	6016	634	150	3614	452.35	0.00%	10.1/sec	41.61	6.80	4227.0
Cart	6015	621	148	3883	445.10	0.00%	10.1/sec	46.00	6.91	4674.0
SignOut	6137	370	293	7885	159.56	0.00%	10.3/sec	53.07	13.08	5278.6
Select_Prod...	6039	194	147	1157	111.92	0.00%	10.1/sec	37.82	7.01	3823.9
Home Page	5907	186	146	1306	100.17	0.00%	9.9/sec	14.05	5.36	1452.0
Signin_Page	6185	196	146	1519	114.71	0.00%	10.4/sec	41.52	7.11	4098.8
Choose_Pet	6137	196	146	3656	125.88	0.00%	10.3/sec	36.90	6.80	3670.6
Product Page	5995	209	147	1348	121.10	0.00%	10.1/sec	48.87	6.12	4978.0
Pet_Specifici...	6113	210	147	2047	147.73	0.00%	10.2/sec	38.04	7.02	3800.8
Signin_Prog...	5968	238	149	2191	178.68	0.00%	10.0/sec	40.77	10.30	4179.6
Confirmation	5998	247	148	1918	178.84	0.00%	10.0/sec	44.32	11.50	4521.2
Billing_Addr...	5988	250	148	21037	320.79	0.02%	10.0/sec	44.79	13.29	4574.2
TOTAL	72498	296	146	21037	284.31	0.00%	121.0/sec	485.81	100.90	4110.7

1. Average Response Times:

Highest: Billing Form (634 ms)

Lowest: Select Product (194 ms)

Medium: Home Page (186 ms), Product Page (209 ms), Pet Specifics (210 ms)

2. Throughput:

Highest: Choose Pet (10.3 requests/sec)

Lowest: Home Page (9.9 requests/sec)

Medium: Most pages are around 10.0 requests/sec

3. Error Rates:

Highest: Billing Address (0.02%)

Spike Test

Latency Report

Latency				
Row Labels	StdDev of Latency	Average of Latency	Max of Latency	Min of Latency
Billing_Address	825.2883597	781.5636	5950	0
Billing_Form	785.2150581	777.1824	3998	154
Cart	843.2342471	826.6011	5940	152
Choose_Pet	858.2041901	824.2099	4301	0
Confirmation_Delivery	782.2796538	773.6018	5925	0
Delivery_Address	853.4419764	850.8327	5932	152
Home Page	771.6424887	807.2404	3941	0
Pet_Speacification	805.4044997	778.423	5905	152
Product Page	775.5704745	778.5195	3927	152
Select_Product	767.8094006	772.0239	3920	151
Signin_Page	831.1394493	845.5109	4111	153
Signin_Progress	858.0239202	888.2784	5922	153
SignOut	782.2837499	766.3075	5951	150
SignOut-0	782.2837499	766.3075	5951	150
SignOut-1	217.4685989	237.7113	1549	150
Grand Total	798.2473542	768.4345	5951	111.26

1. StdDev of Latency:

Highest: Choose_Pet (858.20 ms)

Lowest: SignOut-1 (217.47 ms)

2. Average Latency:

Highest: Signin_Progress (888.28 ms)

Lowest: SignOut-1 (237.71 ms)

3. Max Latency:

Highest: SignOut (5951 ms)

Lowest: Select_Product (3920 ms)

4. Min Latency:

Highest: Billing_Address (154 ms)

Lowest: SignOut-1 (159 ms)

Aggregate Report

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received K...	Sent KB/sec
Cart	524	875	653	2177	2730	3332	152	6098	0.00%	4.0/sec	21.10	2.56
Confirmati...	555	1054	563	2212	2427	3171	159	21034	100.00%	4.2/sec	64.16	2.54
SignOut	478	1037	805	2159	2503	4239	301	6299	0.00%	3.7/sec	19.17	4.47
Billing_Form	488	816	207	2042	2426	3278	154	4182	0.00%	3.8/sec	17.59	2.35
Choose_Pet	543	925	200	2337	2732	4045	151	21033	0.18%	4.2/sec	17.57	2.56
Pet_Speaci...	539	819	201	2131	2422	3287	152	6060	0.19%	4.2/sec	17.76	2.65
Home Page	495	892	674	1948	2291	2966	151	21035	0.40%	3.8/sec	5.45	1.87
Select_Pro...	544	818	196	2116	2392	2994	151	4084	0.00%	4.2/sec	18.19	2.70
Product Pa...	564	836	209	2141	2515	3049	152	4095	0.00%	4.4/sec	24.20	2.44
Signin_Pro...	528	949	686	2321	2654	3016	153	6092	0.19%	4.1/sec	18.99	4.02
Signin_Page	548	900	611	2214	2617	3359	153	4114	0.00%	4.3/sec	19.44	2.71
Billing_Ad...	550	894	196	2110	2649	3154	152	21060	0.36%	4.3/sec	21.32	5.51
Delivery_A...	520	893	240	2147	2514	3263	152	6090	0.19%	4.1/sec	20.42	4.53
TOTAL	6876	900	376	2151	2551	3306	151	21060	8.19%	52.6/sec	281.67	40.22

Response Time Analysis Summary

1. Average Response Times:

Highest: Confirmation (1054 ms)

Lowest: Select Product (818 ms)

2. 95th Percentile Response Times:

Highest: SignOut (4203 ms)

Lowest: Cart (2730 ms)

3. 99th Percentile Response Times:

Highest: SignOut (4239 ms)

Lowest: Home Page (2994 ms)

Summary report

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/...	Sent KB/sec	Avg. Bytes
Cart	524	875	152	6098	906.29	0.00%	4.0/sec	21.10	2.56	5360.0
Confirmation_Delivery	555	1054	159	21034	1201.81	100.00%	4.2/sec	64.16	2.54	15471.1
SignOut	478	1037	301	6299	870.10	0.00%	3.7/sec	19.17	4.47	5353.0
Billing_Form	488	816	154	4182	838.56	0.00%	3.8/sec	17.59	2.35	4780.9
Choose_Pet	543	925	151	21033	1268.38	0.18%	4.2/sec	17.57	2.56	4266.2
Pet_Specification	539	819	152	6060	862.64	0.19%	4.2/sec	17.76	2.65	4356.0
Home Page	495	892	151	21035	1495.85	0.40%	3.8/sec	5.45	1.87	1457.3
Select_Product	544	818	151	4084	827.54	0.00%	4.2/sec	18.19	2.70	4423.0
Product Page	564	836	152	4095	841.28	0.00%	4.4/sec	24.20	2.44	5664.0
Signin_Progress	528	949	153	6092	925.25	0.19%	4.1/sec	18.99	4.02	4734.0
Signin_Page	548	900	153	4114	898.50	0.00%	4.3/sec	19.44	2.71	4654.1
Billing_Address	550	894	152	21060	1496.07	0.36%	4.3/sec	21.32	5.51	5035.7
Delivery_Address	520	893	152	6090	901.68	0.19%	4.1/sec	20.42	4.53	5057.0
TOTAL	6876	900	151	21060	1056.59	8.19%	52.6/sec	281.67	40.22	5486.1

Throughput (Requests per second under different loads):

- Highest Throughput:
Billing Address: 4.3 requests/sec
- Lowest Throughput:
SignOut: 3.7 requests/sec
- Others (Medium):
Most pages around 4.0 to 4.2 requests/sec
TOTAL: 52.6 requests/sec

Error Rates (Error rates and HTTP status code distribution):

- Highest Error Rate:
Confirmation_Delivery: 100.00%
- Lowest Error Rates:
Cart, SignOut, Billing_Form, Home Page, Select_Product, Product Page, Signin_Page, Billing_Address, Delivery_Address: 0.00%
- Others (Medium):
Choose_Pet: 0.18%
Pet_Specification: 0.19%
Signin_Progress: 0.19%
TOTAL: 8.19%

Endurance Test

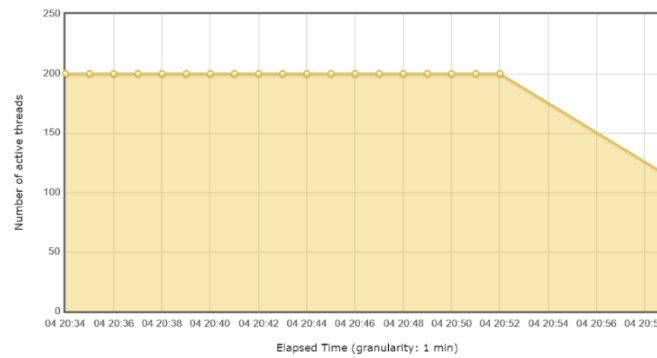
Test Description :

200 User/Sec continuously hits upto 30 mins.

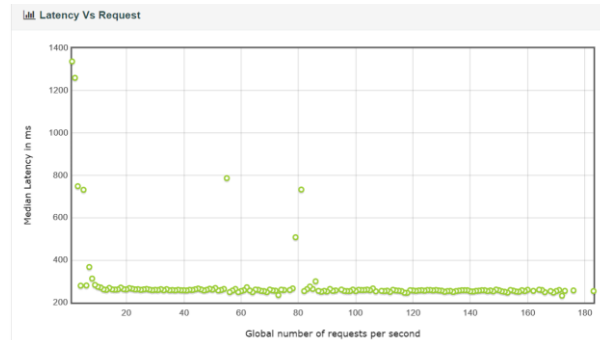
This will be achieved by using synchronizing timer and normal thread group.



This Graph clearly states that 4lakh ms is the average response time during the test.



This graph clearly indicates that 200 active threads continuously hit's the website over the period of time.



In this Latency report maximum hit's by 800. Majorly maintained at 100 Latency.

Volume Test

Test Description :

In this Volume test which conduct by 3 thread groups hits 10 mints. Each thread group ends at duration of each 600secs after another thread group initiated. First thread group started with 100 users. Second thread group started with 300 users. And Final thread group started with 500 users.

Scalability A – 100 Users Aggregate Report, Summary Report and Latency

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received KB/sec	Sent KB/sec
Cart	1033	339	302	314	796	1202	207	2064	0.10%	1.8/sec	9.20	1.12
Billing_Form	1046	348	303	312	747	1232	210	21060	0.10%	1.8/sec	8.30	1.11
SignOut	1010	522	490	520	886	1198	401	2252	0.69%	1.7/sec	8.98	2.09
Signin_Page	1040	283	245	265	729	869	195	1701	0.00%	1.8/sec	8.03	1.12
Choose_Pet	1025	297	244	290	749	943	191	2033	0.10%	1.7/sec	7.26	1.06
Select_Prod...	1041	280	246	264	719	807	185	1910	0.00%	1.8/sec	7.64	1.13
Product Pa...	1111	288	245	270	739	847	190	1256	0.00%	1.9/sec	10.45	1.05
Home Page	1059	278	245	263	730	907	184	1164	0.00%	1.8/sec	2.55	0.88
Pet_Speacif...	1037	282	245	265	729	932	200	950	0.00%	1.8/sec	7.50	1.12
Delivery_A...	1063	317	250	680	762	1031	198	2028	0.09%	1.8/sec	8.92	1.98
Billing_Add...	987	319	249	669	758	1051	184	2589	0.10%	1.7/sec	8.27	2.14
Signin_Pro...	1036	294	248	300	730	994	200	1650	0.00%	1.8/sec	8.15	1.73
TOTAL	12488	320	251	507	742	1002	184	21060	0.10%	21.2/sec	94.98	16.48

Response Time Analysis Summary

- **Average Response Times: - 320 ms**

Highest: Signout (522ms)

Lowest: Home page (278 ms)

- **95th Percentile Response Times: - 742 ms**

Highest: Signout (886ms)

Lowest: Select_product (719 ms)

- **99th Percentile Response Times: - 1002 ms**

Highest: Signout (1198 ms)

Lowest: Home page (807 ms)

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
Cart	1033	339	207	2054	189.41	0.10%	1.8/sec	9.20	1.12	5360.0
Billing_Form	1046	348	210	21060	661.68	0.10%	1.8/sec	8.30	1.11	4779.0
SignOut	1010	522	401	2252	189.72	0.69%	1.7/sec	8.98	2.09	5353.0
Signin_Page	1040	283	195	1701	154.46	0.00%	1.8/sec	8.03	1.12	4653.9
Choose_Pet	1025	297	191	2033	192.41	0.10%	1.7/sec	7.26	1.06	4269.0
Select_Product	1041	280	185	1910	147.58	0.00%	1.8/sec	7.64	1.13	4423.0
Product Page	1111	288	190	1256	152.11	0.00%	1.9/sec	10.45	1.05	5664.0
Home Page	1059	278	184	1164	142.45	0.00%	1.8/sec	2.55	0.88	1452.0
Pet_Speacification	1037	282	200	950	143.92	0.00%	1.8/sec	7.50	1.12	4356.0
Delivery_Address	1063	317	198	2028	196.48	0.09%	1.8/sec	8.92	1.98	5057.0
Billing_Address	987	319	184	2589	220.72	0.10%	1.7/sec	8.27	2.14	5044.0
Signin_Progress	1036	294	200	1650	164.59	0.00%	1.8/sec	8.15	1.73	4734.0
TOTAL	12488	320	184	21060	261.53	0.10%	21.2/sec	94.98	16.48	4593.9

Error Rate

- Total Error rate is 0.10%.
- Max 0.62% (Signout) and Min 0.10% (Cart, Billing Form, etc) Error Percentage

Throughput

- Total Throughput rate is 134.9/sec.
- Thruhtput remains 11.1/sec to 11.3/sec.

Row Labels	StdDev of Latency	Average of Latency	Max of Latency	Min of Latency
Billing_Address	206.0844024	316.0061	2282	184
Billing_Form	138.5557302	321.2017	1638	0
Cart	153.0397524	326.668	2064	207
Choose_Pet	177.3576953	293.7298	1785	191
Delivery_Address	186.399736	312.9972	1781	198
Home Page	142.5224606	278.289	1164	184
Pet_Speacification	143.8518072	281.757	948	200
Product Page	148.2825452	286.5608	1160	189
Select_Product	139.0658687	277.6955	1672	185
Signin_Page	145.7834437	280.8846	1701	195
Signin_Progress	157.1465871	292.2751	1453	200
SignOut	163.6165089	275.9069	1808	204
SignOut-0	163.6165089	275.9069	1808	204
SignOut-1	42.21350828	243.5198	1223	188
Grand Total	155.9551775	290.2989	2282	180.64

- 1. Average Latency:** The overall average latency is approximately 290 ms. The "Cart" step has the highest average latency (327 ms), while "SignOut-1" shows the lowest average latency (244 ms).
- 2. Standard Deviation (StdDev):** The average StdDev is 156 ms, with "SignOut-1" exhibiting the highest variability (207 ms), indicating inconsistent performance.
- 3. Maximum Latency:** All steps have maximum latency peaks around 2282 ms, showing occasional long delays across processes.
- 4. Minimum Latency:** Most steps have minimum latencies between 191 ms and 207 ms, except for "Billing Form" (0 ms).

Scalability B – 300 Users Aggregate Report

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received KB...	Sent KB/sec
Billing_Form	6687	360	232	698	1068	1633	157	4571	0.51%	11.2/sec	52.34	6.98
Cart	6730	356	231	688	1065	1506	160	6959	0.34%	11.3/sec	59.05	7.16
Signin_Page	6751	341	230	643	988	1494	159	3412	0.30%	11.3/sec	51.47	7.18
SignOut	6910	573	457	910	1238	1801	333	6721	0.67%	11.6/sec	60.60	14.13
Choose_Pet	6678	343	230	650	950	1495	158	4141	0.37%	11.2/sec	46.71	6.83
Home_Page	6708	275	222	273	776	1069	157	3772	0.10%	11.3/sec	15.96	5.51
Product Pa...	6643	370	234	694	1084	1757	158	3908	0.72%	11.1/sec	61.65	6.21
Select_Prod...	6802	343	231	642	955	1537	160	5361	0.38%	11.4/sec	49.30	7.31
Pet_Speacif...	6671	354	230	680	1033	1583	167	7696	0.48%	11.2/sec	47.68	7.10
Signin_Pro...	6646	355	233	669	976	1585	167	4628	0.56%	11.2/sec	51.64	10.94
Billing_Add...	6682	349	232	663	952	1540	166	3930	0.46%	11.2/sec	55.38	14.33
Delivery_A...	6594	347	231	651	940	1508	157	3962	0.41%	11.1/sec	54.82	12.15
TOTAL	80502	364	235	677	1010	1584	157	7696	0.44%	134.9/sec	605.63	105.64

Response Time Analysis Summary

- Average Response Times: - 364 ms**
 Highest: Signout (573ms)
 Lowest: Home page (275 ms)
- 95th Percentile Response Times: - 1010 ms**
 Highest: Signout (1238ms)
 Lowest: Home page (776 ms)
- 99th Percentile Response Times: - 1584 ms**
 Highest: Signout (1801 ms)
 Lowest: Home page (1069 ms)

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
Billing_Form	6687	360	157	4571	318.79	0.51%	11.2/sec	52.34	6.98	4781.0
Cart	6730	356	160	6959	318.11	0.34%	11.3/sec	59.05	7.16	5360.0
Signin_Page	6751	341	159	3412	285.54	0.30%	11.3/sec	51.47	7.18	4653.9
SignOut	6910	573	333	6721	325.33	0.67%	11.6/sec	60.60	14.13	5353.0
Choose_Pet	6678	343	158	4141	290.58	0.37%	11.2/sec	46.71	6.83	4269.0
Home Page	6708	275	157	3772	202.31	0.10%	11.3/sec	15.96	5.51	1452.0
Product Page	6643	370	158	3908	334.34	0.72%	11.1/sec	61.65	6.21	5664.0
Select_Product	6802	343	160	5361	291.22	0.38%	11.4/sec	49.30	7.31	4423.0
Pet_Specifica...	6671	354	167	7696	325.13	0.48%	11.2/sec	47.68	7.10	4356.0
Signin_Progress	6646	355	167	4628	308.59	0.56%	11.2/sec	51.64	10.94	4734.0
Billing_Address	6682	349	166	3930	293.18	0.46%	11.2/sec	55.38	14.33	5044.0
Delivery_Addr...	6594	347	157	3962	289.66	0.41%	11.1/sec	54.82	12.15	5057.0
TOTAL	80502	364	157	7696	307.97	0.44%	134.9/sec	605.63	105.64	4595.9

Error Rate

- Total Error rate is 0.44%.
- Max 0.72% (Product_page) and Min 0.10% (Home Page) Error Percentage

Throughput

1. Total Throughput rate is 134.9/sec.
2. Throughtput remains 11.1/sec to 11.3/sec.

Row Labels	StdDev of Latency	Average of Latency	Max of Latency	Min of Latency
Billing_Address	237.9721387	301.2312	3930	166
Billing_Form	263.3216232	309.5932	4313	157
Cart	257.1814676	307.1105	6120	160
Choose_Pet	239.1948552	296.3672	4141	158
Delivery_Address	237.7663615	298.8274	3843	157
Home Page	202.3267344	275.213	3772	157
Pet_Speacification	267.7514156	305.428	7696	167
Product Page	227.3583896	292.4158	3342	149
Select_Product	234.4605805	292.3434	5361	160
Signin_Page	227.5891009	294.1686	3025	157
Signin_Progress	254.8930514	303.9231	4397	165
SignOut	125.9905717	246.1644	2035	163
SignOut-0	125.9905717	246.1644	2035	163
SignOut-1	226.9454939	280.6219	5111	167
Grand Total	228.0657277	289.0218	7696	149

1. **Average Latency:** The overall average latency is approximately 289 ms. The "Billing_Form" step has the highest average latency (309 ms), while "SignOut" shows the lowest average latency (246 ms).
2. **Standard Deviation (StdDev):** The average StdDev is 228 ms, with "Pet_Speacification" exhibiting the highest variability (268 ms), indicating inconsistent performance.
3. **Maximum Latency:** All steps have maximum latency peaks around 7696 ms, showing occasional long delays across processes.
4. **Minimum Latency:** Most steps have minimum latencies between 157 ms and 166 ms, except for "Product Page" (149 ms).

Scalability C – 500 Users Aggregate Report

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
Billing_Form	12564	294	152	27536	815.23	0.99%	21.0/sec	98.13	13.09	4780.8
Cart	12448	298	153	23371	846.19	0.96%	20.8/sec	109.00	13.22	5360.0
SignOut	12316	581	301	31131	1659.15	1.23%	20.6/sec	107.74	25.12	5353.0
Product Page	12230	302	149	26184	873.47	1.03%	20.5/sec	113.22	11.41	5664.0
Home Page	12533	287	150	24076	810.92	0.94%	21.0/sec	29.75	10.27	1452.0
Choose_Pet	12389	297	153	21338	832.72	1.04%	20.7/sec	86.45	12.64	4269.0
Select_Product	12593	287	152	14836	764.96	0.94%	21.1/sec	91.07	13.51	4423.0
Signin_Page	12421	300	150	19684	840.48	1.13%	20.8/sec	94.50	13.18	4654.0
Pet_Specifica...	12520	309	151	17532	862.81	1.07%	21.0/sec	89.21	13.29	4356.0
Delivery_Addr...	12513	329	150	21111	958.08	1.42%	21.0/sec	103.63	22.97	5056.8
Billing_Address	12318	329	153	22128	959.26	1.37%	20.7/sec	101.73	26.32	5044.0
Signin_Progress	12456	342	153	27346	1082.73	1.48%	20.9/sec	96.57	20.46	4733.8
TOTAL	149301	329	149	31131	972.19	1.13%	249.7/sec	1119.43	195.17	4591.3

Error Rate

- Total Error rate is 1.13%.
- Max 1.42% (Deliver_Address) and Min 0.94% (Home Page) Error Percentage

Throughput

1. Total Throughput rate is 249.7/sec.
2. Throughtput remains 20.7/sec to 21.1/sec.

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received KB...	Sent KB/sec
Billing_Form	12564	294	198	240	443	1802	152	27536	0.99%	21.0/sec	98.13	13.09
Cart	12448	298	198	244	523	1802	153	23371	0.96%	20.8/sec	109.00	13.22
SignOut	12316	581	391	456	748	6654	301	31131	1.23%	20.6/sec	107.74	25.12
Product Pa...	12230	302	197	243	527	2321	149	26184	1.03%	20.5/sec	113.22	11.41
Home Page	12533	287	195	235	456	1779	150	24076	0.94%	21.0/sec	29.75	10.27
Choose_Pet	12389	297	196	238	508	2258	153	21338	1.04%	20.7/sec	86.45	12.64
Select_Prod...	12593	287	196	238	493	1553	152	14836	0.94%	21.1/sec	91.07	13.51
Signin_Page	12421	300	196	237	514	3249	150	19684	1.13%	20.8/sec	94.50	13.18
Pet_Speacif...	12520	309	196	248	575	3053	151	17532	1.07%	21.0/sec	89.21	13.29
Delivery_A...	12513	329	197	259	592	4925	150	21111	1.42%	21.0/sec	103.63	22.97
Billing_Add...	12318	329	197	258	598	4971	153	22128	1.37%	20.7/sec	101.73	26.32
Signin_Pro...	12456	342	198	253	590	5962	153	27346	1.48%	20.9/sec	96.57	20.46
TOTAL	149301	329	199	392	563	2764	149	31131	1.13%	249.7/sec	1119.43	195.17

Response Time Analysis Summary

1. **Average Response Times: - 329 ms**
 Highest: Signout (581ms)
 Lowest: Home page (287 ms)
2. **95th Percentile Response Times: - 563 ms**
 Highest: Signout (748 ms)
 Lowest: Billing Form (443 ms)
3. **99th Percentile Response Times: - 2764 ms**
 Highest: Signout (6654 ms)
 Lowest: Billing Form (1082 ms)

Row Labels	StdDev of Latency	Average of Latency	Max of Latency	Min of Latency
Billing_Address	926.4235459	322.3942198	21138	150
Billing_Form	789.7650211	285.9694365	27535	0
Cart	791.187882	287.2596401	22313	150
Choose_Pet	814.5787502	291.1051739	21338	152
Delivery_Address	932.1998242	321.5964996	21107	0
Home Page	810.9564318	287.5368228	24076	150
Pet_Speacification	857.9954779	302.9674121	17532	151
Product Page	824.103855	289.3792314	19445	148
Select_Product	748.9573829	281.0822679	13537	152
Signin_Page	828.5301259	294.0096611	19684	150
Signin_Progress	1062.426254	335.4605812	27346	0
SignOut	802.2531688	283.9984573	15272	149
SignOut-0	802.2531688	283.9984573	15272	149
SignOut-1	746.2261167	266.8433745	22787	150
Grand Total	842.5387856	295.2776701	27535	117.97

- 1. Average Latency:** The overall average latency is approximately 295 ms. The "Signin_Process" step has the highest average latency (336 ms), while "SignOut" shows the lowest average latency (267 ms).
- 2. Standard Deviation (StdDev):** The average StdDev is 843 ms, with "Signin_Process" exhibiting the highest variability (268 ms).
- 3. Maximum Latency:** All steps have maximum latency peaks around 27535 ms, showing occasional long delays across processes.
- 4. Minimum Latency:** Most steps have minimum latencies between 148 ms and 152 ms.

Key Findings

a) Load Test

After analysing load test signin-page takes higher latency duration to connect. Have to focus on this page.

b) Stress test

After analysing stress test error occurs in billing form page 0.02%. But higher latency occurs signin page.

c) Spike Test

After analysing spike test major impact created by confirmation page 100% error. Signin-progress takes maximum duration of latency.

d) Volume Test

After analysing volume test higher error occurs. Home page have very good stability in performance testing. Overall very minimal error occurs.

e) Endurance Test

During the endurance test application stability is very good. And nill errors. Throughput transactions maintained at 2 secs for overall period.

Conclusions

The website is operating extremely steadily overall after all the tests and reports have been examined. The confirmation page and the signout page need to be given more attention. Excellent performance on the home page.